



ELA: Recursive Linguistic Elements & the Context–Pattern Duality

An AI-Assisted Framework for Authentic English Comprehension — Where does language end and pattern begin?

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Problem & Motivation

Language-learning platforms rely on **pre-simplified materials**. Authentic English — films, podcasts, news — is linguistically rich but cognitively inaccessible without structural guidance. Simplification destroys the input; learners never encounter real language.

Hypothesis: Deterministic NLP and lightweight generative AI can *explain* authentic input without altering it — producing hierarchical annotations that scaffold meaning while preserving full linguistic complexity.

- How can a hybrid NLP + LLM pipeline produce accurate pedagogical annotations from authentic media?
- What data model captures recursive linguistic relations while remaining human-interpretable?
- Can bilingual visualisation reduce cognitive load?

The RLE Data Model

Core innovation: every sentence is a **nested JSON hierarchy** at three levels — Sentence → Phrase → Word — validated against a dual-schema contract before persistence.

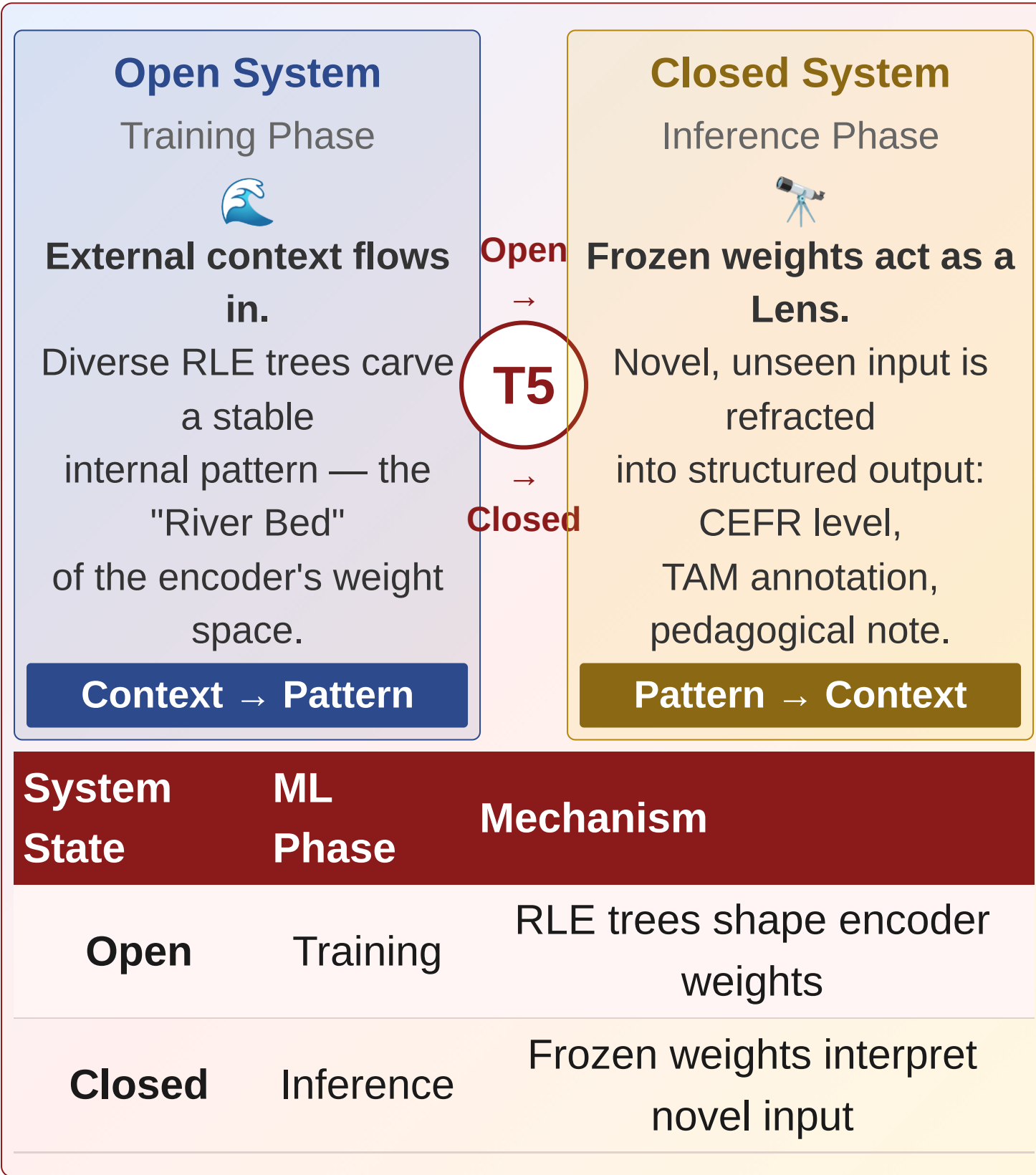
Sentence
└─ Phrase (NP / VP / PP / ADJP)
└─ Word
└─ translation
└─ CEFR level (A1–C2)
└─ POS & IPA
└─ TAM (Tense–Aspect–Mood)
└─ note (T5-generated)

The New Physics of Context–Pattern Interaction

**"In open systems, context forms patterns.
In closed systems, patterns form context."**

— Principle discovered empirically during T5 fine-tuning

The Duality Loop Diagram



The Key Insight: Placeholder-Driven Learning

Fine-tuning T5 on raw linguistic notation **failed to generalise**. The breakthrough was replacing lexical tokens with structural placeholders:

✗ Raw: "She gave him a book" → [VP gave][NP him]

✓ Template: [SUBJ][V-PAST][IOBJ][DET][NOUN] → [TAM: past-simple][VOICE: active]

The abstract grammatical **pattern** — not the words — became the unit of learning. The model generalises to novel

Performance Metrics

78.3%

CEFR Classification Accuracy
XGBoost + Random Forest + Logistic Regression
3,000-sentence CEFR-balanced corpus

0.81

Note Relevance (ROUGE-L)
Fine-tuned T5 model evaluation on held-out RLE annotation set

8.2%

ASR Word Error Rate
Whisper base model clean speech test set

Annotation Backend Comparison

Approach	Quality	Cost	Offline
GPT-4o	High	\$\$\$	✗
Rules only	Medium	Free	✓
T5 fine-tuned	High	Free	✓
XGBoost CEFR	78.3%	Free	✓

T5 fine-tuning achieves GPT-quality annotation at zero inference cost, running entirely on-device.

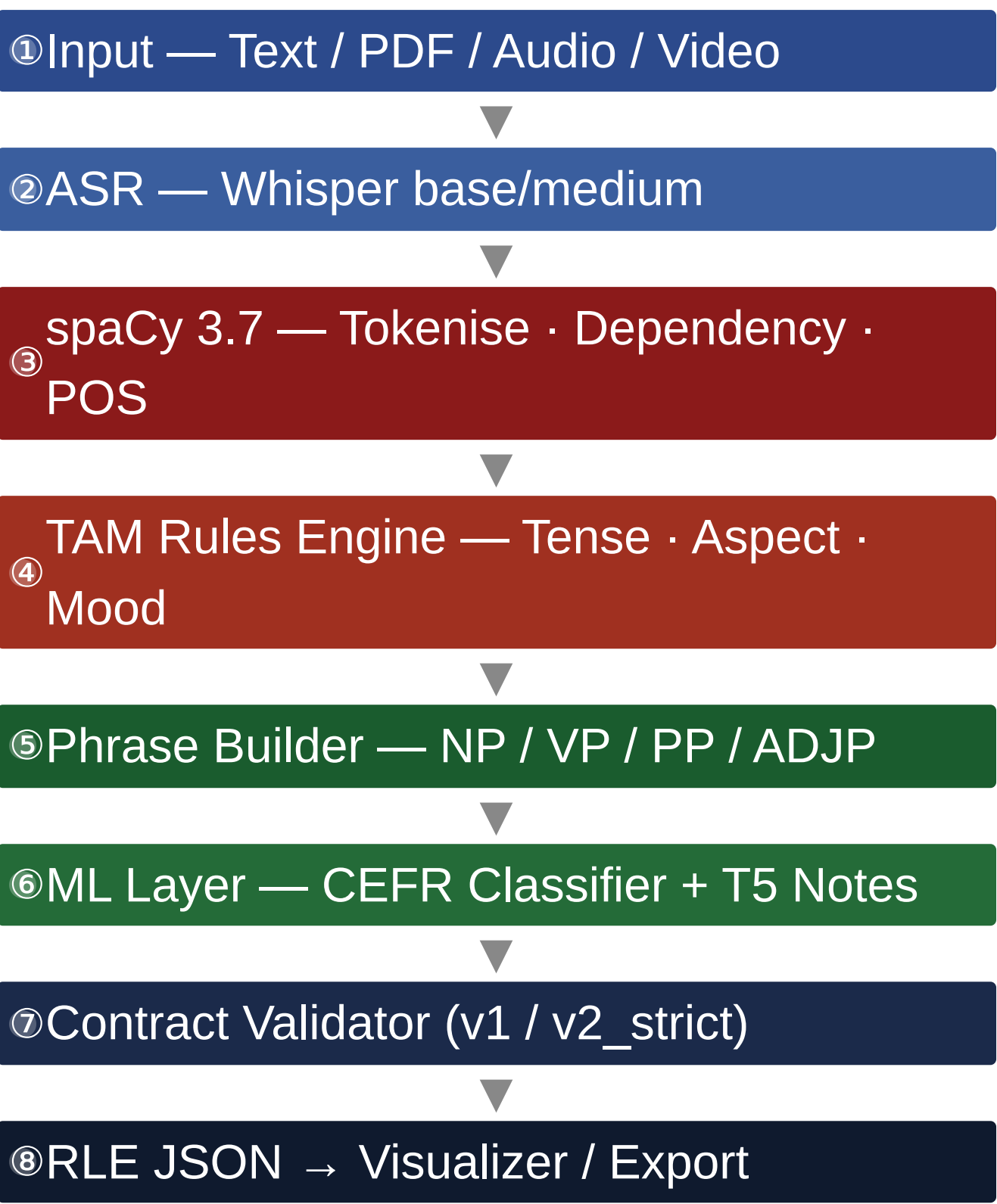
System Validation

- O1 (RLE Generation):** spaCy + TAM engine produces schema-valid RLE for all test sentences ✓
- O2 (ML Enrichment):** CEFR 78.3%; T5 ROUGE-L 0.81 ✓
- O3 (UI & Export):** Linguistic Visualizer renders full RLE tree; JSON/CSV export verified ✓
- O4 (Deployment):** Android PWA install — fully client-side, no cloud dependency ✓

Conclusions

Schema is **language-agnostic**.
Current focus: English → Russian.
Pluggable translator backends
support additional target languages.

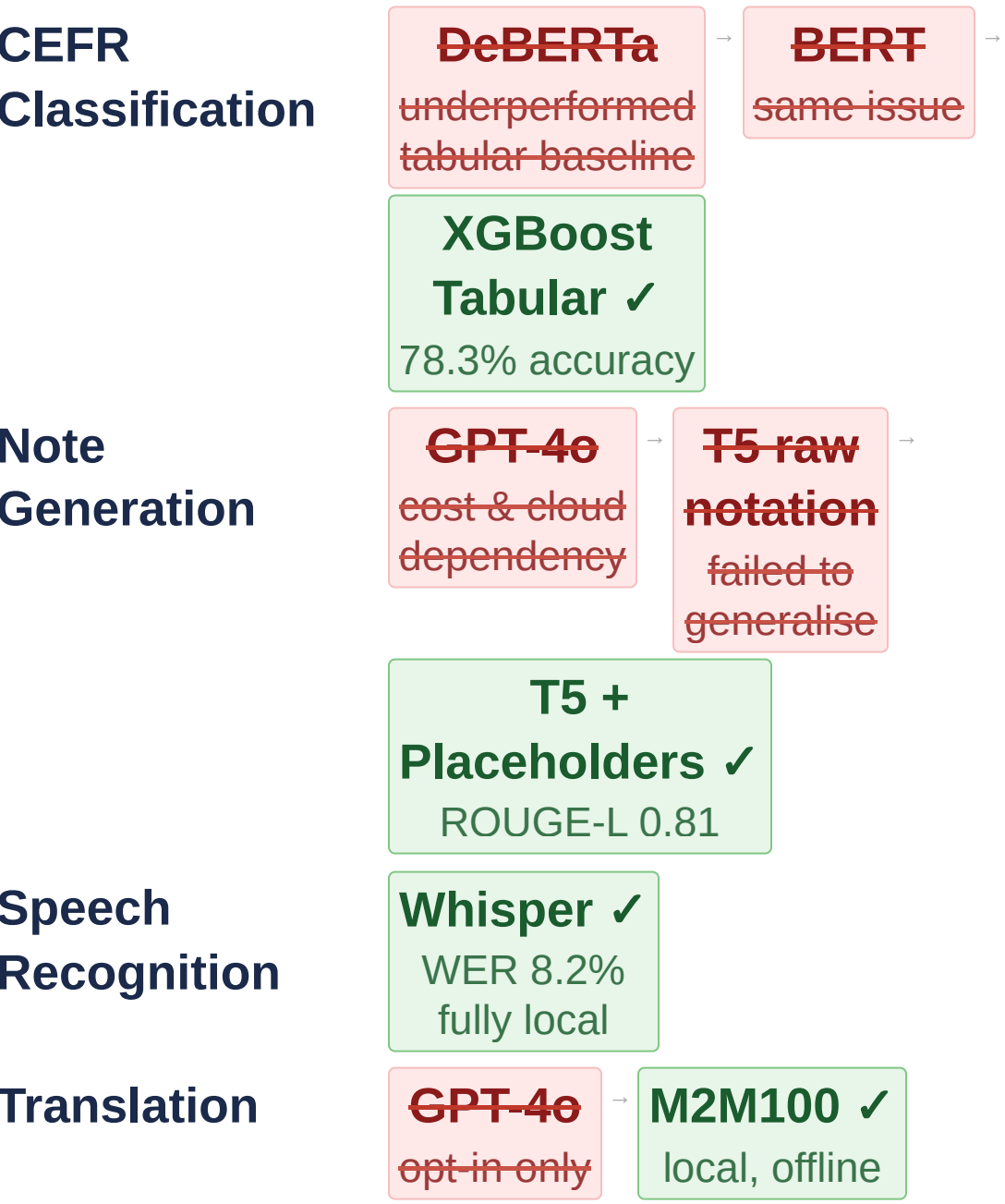
Analysis Pipeline



Deployed as **Progressive Web App (PWA)**
— Android-installable, fully client-side. No
cloud dependency. No data leaves the device.

Research Journey: What We Tried

Each component went through multiple
candidate approaches before the final solution
was selected:



vocabulary because the pattern is
invariant to lexical substitution.

*This is closed-system behaviour: the internal
pattern imposes structure on any incoming
context — regardless of whether that context
was seen during training.*

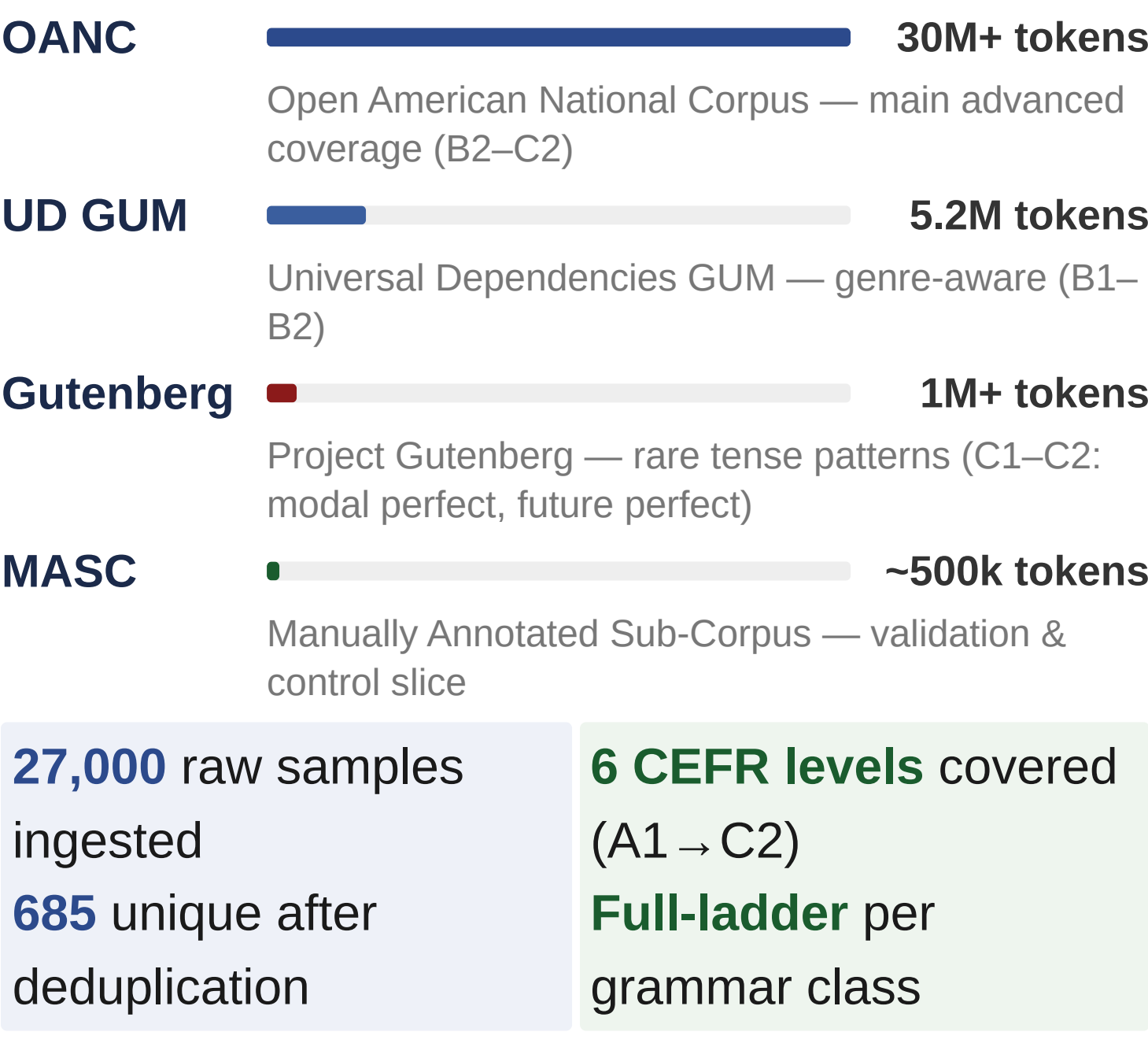
Implication for Agentic AI

An LLM agent navigating the real world is a **closed system** operating in an **open environment**. Its
trained patterns are fixed; the world's context is
not. The duality principle predicts exactly where
agents succeed (stable structural patterns) and fail
(distribution shift beyond the trained pattern's
boundary).

*Understanding training as pattern-carving and
inference as pattern-imposing offers a new
lens for agent reliability research.*

Training Data & Corpus Scale

Four open-access corpora merged into a CEFR-
labelled full-ladder dataset (A1 → C2):



Live Demo

www.el-a.uk

Try it in your browser

- Explainable AI annotation makes authentic English **pedagogically accessible without simplification**.
- The RLE schema is a viable, human-interpretable model of recursive linguistic structure.
- The **Context–Pattern Duality**, observed empirically during T5 fine-tuning, generalises beyond NLP to any learning system that transitions from open-world training to closed-system inference.
- Placeholder-driven T5 fine-tuning generalises to novel vocabulary because the abstract structural pattern is lexically invariant.

Future Directions

- **Tauri 2 Desktop App:** Native offline package (Linux .deb / Windows) with bundled ML models — no browser required, full hardware access for on-device inference.
- **Translation Quality Gates:** Round-trip ChrF scorer, length-ratio guards, n-gram repetition penalty, and a human review queue wired into the M2M100 pipeline.
- **M2M100 LoRA Fine-Tuning:** Domain-specific refinement of the translation model on accepted English → Russian pairs from the review queue.
- **Additional target languages:** Pluggable translator backends support any language beyond the current English → Russian focus.
- **Real-time streaming ASR:** Live audio analysis for conversation practice via Whisper streaming mode.

References

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